Wearable-Based Digital Biomarkers for Parkinson's Disease

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Behavioral Health Informatics D

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1 Introduction

Parkinson's disease (PD) is a progressive neurodegenerative disorder characterized by motor symptoms such as tremor, rigidity, and bradykinesia Shawen et al. (2021). Currently, diagnosis and severity assessment rely primarily on periodic clinical examinations and subjective rating scales, such as the MDS-UPDRS or the Hoehn and Yahr (H&Y) scale Mittal et al. (2021). However, these evaluations can be infrequent and subject to inter-rater variability Shawen et al. (2021).

In this context, wearable technologies, such as smartwatches, provide a powerful alternative by enabling continuous physiological data collection in real-world settings Shawen et al.

[0]Project repository on GitHub: `https://github.com/PaoloFabbri8/Behavioral-Health-Informatic-Project.git`

(2021). Recent literature highlights that the use of inertial sensors (accelerometers and gyroscopes) combined with machine learning algorithms can achieve accuracies exceeding 85% in symptom detection El Maachi et al. (2021). While accelerometers are considered the standard for characterizing human activity, the inclusion of gyroscopes can significantly improve bradykinesia detection Shawen et al. (2021).

Nevertheless, a critical trade-off exists between data complexity and the computational resources of wearable devices, such as battery life Shawen et al. (2021). Moreover, PD diagnosis is complex, requiring the assessment of both motor and non-motor symptoms El Maachi et al. (2021). Many studies have focused exclusively on movement signals, but integrating multimodal data,including clinical questionnaires and demographic information, may provide a more comprehensive view and increase predictive model reliability Mittal et al. (2021).

This project aims to address this gap by analyzing which digital biomarkers, derived from both sensor and clinical data from the PADS dataset (Parkinson's Disease Smartwatch Dataset), are most effective in identifying patients at risk of developing the disease.

1.1 Research Question

Starting from an initial exploratory phase of the dataset and a preliminary analysis of its composition, it was possible to identify three main research questions guiding this work.

Following this, the three research questions were developed:

- **RQ1.** Can machine learning distinguish Parkinson disease from other movement disorders using smartwatch sensor data?
- **RQ2.** Which smartwatch-derived digital biomarkers are most predictive of Parkinson disease?
- **RQ3.** Do smartwatch-based digital biomarkers improve early screening for Parkinson disease compared to self-reported symptoms?

The analysis was structured on two complementary levels. On the one hand, an in-depth analysis of the raw data was conducted in order to highlight relevant patterns, anomalies, and distributions. On the other hand, considerable attention was devoted to descriptive statistics and feature engineering techniques, with the aim of outlining a clearer profile of the research and what was intended to be effectively developed and built.

2 Exploratory Data Analysis (EDA)

The initial phase of the data analysis focused on loading and integrating the three main data sources available: questionnaire data, time-series data collected from smartwatch-based tests, and patients' demographic information.

We began by examining the demographic dataset, with particular attention to missing values. Overall, the data quality was high, with only one variable, `appearance_in_first_grade_kinship`, containing missing values. This variable was subsequently excluded from the analysis, as it was not considered relevant for predictive purposes.

The dataset encompasses six distinct medical conditions, with Parkinson's Disease (PD) representing the majority class (58.8%), followed by healthy controls (16.8%) and other movement disorders (12.8%). Less frequent conditions include Essential Tremor (6.0%), Atypical Parkinsonism (3.2%), and Multiple Sclerosis (2.3%).

This highlights the presence of class imbalance, with PD being substantially overrepresented compared to rarer conditions, which together account for only a small fraction of the sample.

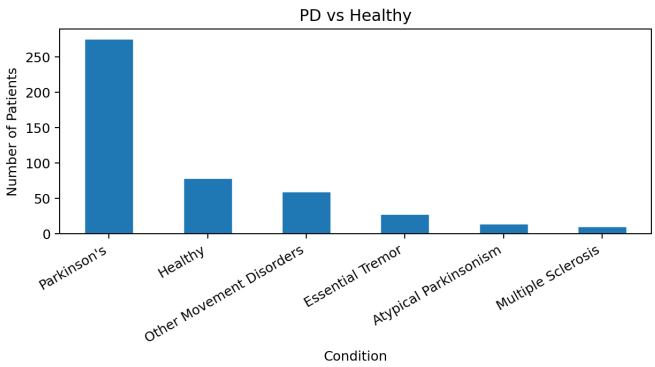

Figure 1: *Barplot of the number of patients with PD vs Healthy*

An ANOVA test was then conducted to assess differences in age across groups, revealing a non-uniform distribution ($p = 0.00117$), with the Parkinson's cohort showing one of the highest median ages. This pattern is likely attributable to the progressive nature of the disease; early symptoms are often subtle or misidentified, leading to a formal diagnosis only in later life as neurodegeneration advances.

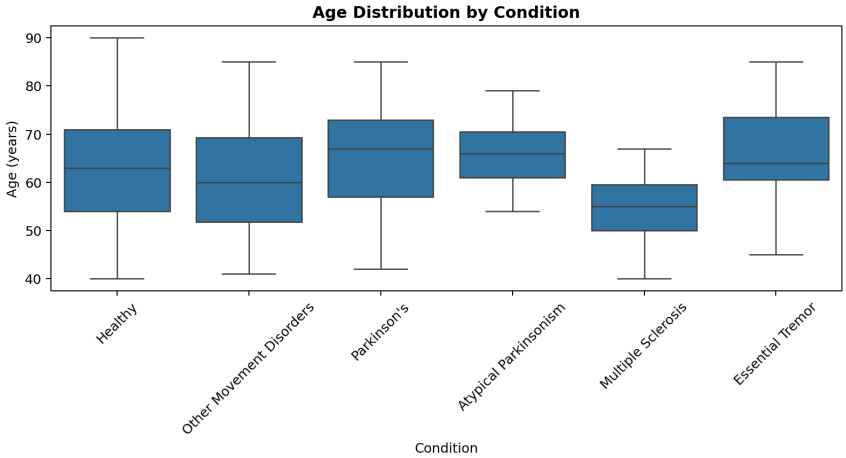

Figure 2: *Boxplot of the age distribution by condition*

Subsequently, we focused on the gender distribution. While women generally exhibit higher life expectancy, the marked male predominance in the Parkinson's cohort ($p < 0.001$) suggests that environmental and occupational factors may play a decisive role. Given that the sample consists of individuals over the age of 50, this gender gap could be linked to historical differences in professional exposure. Men of this generation were more likely to be employed in strenuous manual labor or industries involving exposure to neurotoxic substances (such as pesticides or industrial chemicals), which are recognized risk factors for the development of Parkinson's disease and essential tremors.

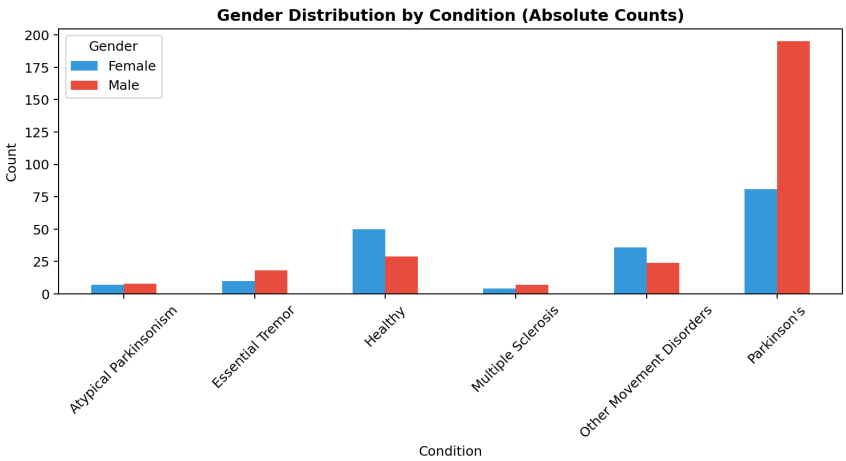

Figure 3: *Barplot of the gender distribution by condition*

The analysis of body weight across the different conditions yielded a statistically significant result through a one-way ANOVA test ($F = 2.439, p = 0.0337$). While the p-value falls below the standard threshold of 0.05, the significance is relatively marginal compared to other demographic variables. The higher body weight observed in Parkinson's patients could likely be attributed to the motor challenges associated with the condition, such as tremors and mobility difficulties. These symptoms often hinder regular physical activity,

leading patients toward a more sedentary lifestyle, which in turn contributes to weight gain.

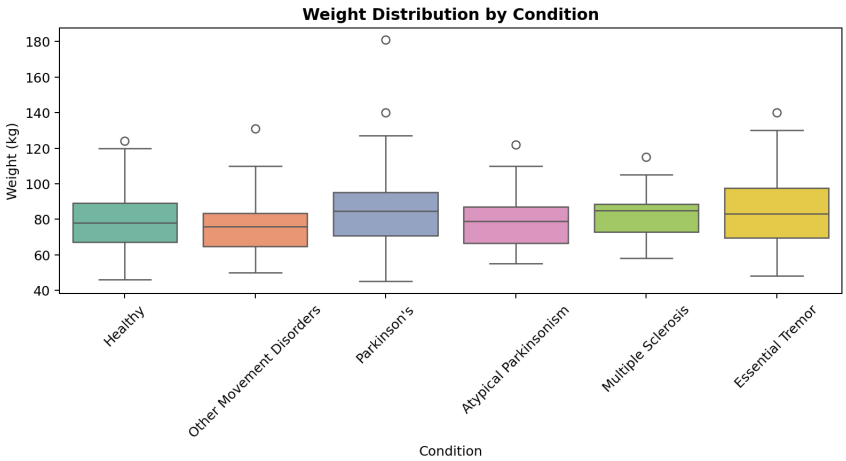

Figure 4: *Barplots of weight distibution by condition*

Following the initial demographic and distribution analysis, we focused on the time-series data collected via smartwatches. To identify which digital biomarkers exhibit the most significant discriminatory power between PD and Non-PD subjects, a Welch's t-test was performed. This univariate analysis, summarized in Table 1, serves to evaluate the mean differences in motion and frequency features, providing a statistical baseline for the subsequent development of predictive models.

The results indicate that only two features remain statistically significant at the $p < 0.05$ level: tremor band power (4-6 Hz) ($p = 0.008$) and mean accelerometer magnitude ($p = 0.044$). Interestingly, the tremor band power is higher in the PD group (0.0017 vs 0.0013), confirming the presence of pathological rest tremor. Conversely, the acceleration mean is lower in PD patients, which may be an objective indicator of bradykinesia (slowness of movement). However, it should be noted that after applying the False Discovery Rate (FDR) correction, these features no longer reach the significance threshold ($p_fdr > 0.05$). This suggests that while these biomarkers show promising trends, the high overlap between PD and Non-PD populations makes individual feature-based classification challenging without more complex machine learning approaches.

Table 1: *Welch's t-test Results: Parkinson's (PD) vs. Non-Parkinson's (Non-PD)*

Feature	Mean PD	Mean Non-PD	t-stat	p-value	p-FDR (BH)	Signif.
tremor_band_4_6Hz	0.0017	0.0013	2.663	0.008	0.054	ns
acc_mag_mean	0.1232	0.1302	-2.011	0.044	0.155	ns
acc_mag_rms	0.1533	0.1606	-1.787	0.074	0.172	ns
gyro_mag_std	0.5533	0.5727	-1.452	0.147	0.256	ns
acc_mag_std	0.0864	0.0889	-1.124	0.261	0.366	ns
dominant_freq_hz	0.7384	0.7622	-0.653	0.514	0.599	ns
gyro_mag_mean	0.7234	0.7326	-0.460	0.645	0.645	ns

BH: Benjamini-Hochberg correction; ns: not significant after FDR correction ($p > 0.05$).

The same statistical approach was applied to the survey responses. In this case, the vast majority of items demonstrated significant differences between the two groups, maintaining their significance even after FDR correction. This high level of discrimination is likely due to the nature of the questions, which are specifically designed to capture early symptoms and pathological signs. Unlike the raw sensor data from smartwatches, which can be influenced by daily variability, the structured survey responses act as a direct clinical proxy for the disease state, thus providing a clearer separation between the Parkinson's and Non-Parkinson's cohorts.

Table 2: *Full Per-Item Significance Summary: Parkinson's vs. Non-Parkinson's*

Item ID	n_PD	n_Non-PD	p-value	p-FDR (BH)	Signif. (FDR)
01	276	193	1.96e-10	1.47e-09	
02	276	193	2.93e-17	8.80e-16	
03	276	193	2.55e-02	3.33e-02	
04	276	193	3.96e-01	4.24e-01	ns
05	276	193	7.61e-09	3.81e-08	
06	276	193	1.00e+00	1.00e+00	ns
07	276	193	1.33e-06	5.69e-06	
08	276	193	8.93e-14	1.34e-12	
09	276	193	4.15e-10	2.49e-09	
10	276	193	1.11e-03	2.08e-03	
11	276	193	2.77e-02	3.41e-02	
12	276	193	2.61e-05	8.71e-05	
13	276	193	2.84e-02	3.41e-02	
14	276	193	8.81e-05	2.40e-04	
15	276	193	2.30e-04	4.93e-04	
16	276	193	4.85e-02	5.60e-02	ns
17	276	193	5.79e-01	5.99e-01	ns
18	276	193	2.33e-02	3.33e-02	
19	276	193	1.78e-04	4.11e-04	
20	276	193	9.15e-03	1.53e-02	
21	276	193	6.93e-04	1.39e-03	
22	276	193	4.10e-05	1.23e-04	
23	276	193	2.29e-01	2.55e-01	ns
24	276	193	1.44e-05	5.41e-05	
25	276	193	5.30e-12	5.30e-11	
26	276	193	9.94e-05	2.48e-04	
27	276	193	2.48e-02	3.33e-02	
28	276	193	1.99e-03	3.51e-03	
29	276	193	1.34e-02	2.01e-02	
30	276	193	1.31e-02	2.01e-02	

BH: Benjamini-Hochberg correction; ns: not significant after FDR correction ($p > 0.05$).

3 Refining the Plan

To develop a robust predictive framework, we implemented a multi-stage feature engineering pipeline.

3.1 Data Load and Merging

The integration process involved merging three heterogeneous data sources (sensor data, questionnaires, and demographic information) using a unique subject identifier in order to construct a comprehensive patient-level dataset.

First, the time series derived from accelerometer and gyroscope sensors were subjected to a structured preprocessing and feature extraction pipeline. An integrated dataset was constructed by linking each motion file to patient information and the corresponding clinical condition (Parkinson's disease / non-Parkinson's) through merge operations with the patient table. The raw signals, acquired along the three axes, were converted into numerical format, handling any invalid values as missing data. Subsequently, the observations were temporally ordered, and missing values were imputed using linear interpolation, complemented by forward and backward filling techniques.

As suggested in the dataset "Usage Notes", the first 0.5 seconds of each recording were removed because the smartwatches produced a vibration notification at the beginning. Signal denoising was then performed by applying a low-pass Butterworth filter (cutoff at 20 Hz) to improve the robustness of time-domain analyses. From the filtered signals, statistical features were extracted for each axis and for the overall magnitude, including mean, standard deviation, root mean square, and range. In addition, frequency-domain features were derived from the accelerometer magnitude, such as dominant frequency, total power, and power within the tremor-related band (4–6 Hz), obtained through a dedicated band-pass filter.

Finally, all extracted features were aggregated at the file level, producing a structured tabular representation saved in CSV format, ready to be merged with the other datasets.

Second, demographic variables such as age, gender, height, and weight were standardized and included to account for physiological variability across individuals.

Finally, binary responses to 30 clinical questionnaire items were transformed into a structured subject-by-feature format, providing a subjective proxy for disease-related symptoms.

The final dataset was constructed by integrating these three data sources through a structured multi-step pipeline. Initially, sensor-derived features were aggregated to obtain a patient-level representation. This was achieved by grouping observations by `subject_id` and condition, and computing the mean and standard deviation for each numerical feature,

excluding non-informative metadata. This transformation reduces dimensionality while preserving intra-patient variability.

To ensure consistency across datasets, a normalization procedure was applied to the patient identifier (`subject_id`), addressing discrepancies in formatting.

The aggregated dataset was then enriched through a multimodal integration step, combining demographic information and questionnaire responses. The latter were reshaped into a subject-by-item format, with one column per question.

Finally, a binary target variable was defined based on the clinical condition: observations were labeled as positive if the condition contained the term "parkinson", and negative otherwise.

3.2 Train Test Splitting and feature selection

A stratified train/test split was performed at the patient level (80% training, 20% test) to ensure a realistic evaluation and prevent data leakage. Splitting by (`subject_id`) guarantees that all observations from the same patient are assigned exclusively to either the training or test set.

Analyzing the data and observing the results of feature selection, we noticed that when merging the three datasets (sensors, demographics, and questionnaires), questionnaire items tended to dominate as the most predictive features. This is likely because the questionnaire responses already serve as an initial screening for the disease, confirming the presence of symptoms that are strongly associated with Parkinson's.

This observation led us to reconsider the research question and refine it to explicitly investigate the differences in predictive power across data modalities. Specifically, we aim to understand whether features derived from sensor-based simulations can achieve comparable predictive performance to self-reported questionnaire items, and how combining both types of information can improve overall disease detection.

- **Updated Research Question (RQ3):** To what extent can smartwatch-based sensor features complement or replicate the predictive power of self-reported questionnaire data for early detection of Parkinson's disease, and how can the combination of both improve predictive performance?

To investigate the relative contribution of different data modalities, three patient-level datasets were created:

- **Dataset A:** sensor-derived features combined with demographic information
- **Dataset B:** questionnaire responses combined with demographic information

- **Dataset C:** combined sensor, questionnaire, and demographic features

We then applied a leakage-safe feature selection procedure on the training set to identify the most informative predictors for the model in all three datasets. First, features with more than 20% missing values were removed, and remaining missing entries were imputed using median values estimated exclusively from the training set. Next, multicollinearity was addressed by removing highly correlated features (correlation > 0.95), retaining those with greater discriminative power.

Subsequently, a bimodal statistical ranking was performed by combining ANOVA F-test and Mutual Information. Features were ranked according to a composite score derived from both metrics, and the top 25 features were selected for the final model.

All thresholds, imputation values, and statistical measures were computed exclusively on the training set and then applied to the test set, ensuring a fully leakage-free pipeline and an unbiased evaluation of model performance.

4 Contributions

This project was entirely carried out independently. Throughout the process, I made use of AI-based tools such as ChatGPT and Gemini primarily to improve the clarity and quality of the written report, as well as to support the search for relevant materials and useful references. Additionally, I used Copilot as a coding assistant to help streamline the development process, particularly for writing code and organizing the project structure. All final decisions, implementations, and interpretations remain my own.

5 References

El Maachi, M. et al. (2021). Deep 1d-convnet for accurate parkinson disease detection and severity prediction from gait. *Computers in Biology and Medicine.*

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